

Semester	B.E. Semester VII– Electronics and Computer Science
Subject	DAV Laboratory
Laboratory Professor	Prof. Uma Jaishankar

Student Name	Jainam Jain
Roll Number	23108B0084
Team Members	Jainam Jain, Vrushan Patil, Dhanush Chowke, Aditya Tambe, Vivek Jaiswal

Mini Project	Socio-Economic & Demographic Analysis of Indian Festivals
Problem Statement	India exhibits immense cultural and regional diversity, with hundreds of festivals celebrated across states, communities and calendar months. Unstructured information leads to biased representations and missing comparative insight. The project must (1) collect an authentic multi-faith festival dataset with zero synthetic records and exact source URLs, (2) build and evaluate ML models, (3) perform rigorous statistical testing and EDA, (4) publish a public data-visualisation blog with 40+ external interactions, and (5) produce a synthesis video.
Resources	Software: Python 3.12, Windows 11, ffmpeg, Windows SAPI (System.Speech) for narration, Microsoft Word (PDF export), MS Edge + Playwright. Libraries: pandas, numpy, scipy, statsmodels, scikit-learn, joblib, matplotlib, plotly, requests, BeautifulSoup, xlrd, markdown. Data sources (all real, cited per record): Reserve Bank of India holiday matrix 2024-26 (rbi.org.in); Census of India 2011 Table C-01 (censusindia.gov.in); Wikipedia infoboxes via MediaWiki API + Govt. portals (utsav.gov.in, soreng.nic.in, dtahills.mn.gov.in, megtourism.gov.in); CAIT (cait.in) and verified news for spending; NRF, gov.cn, GASTAT, muenchen.de for global analogues; DataMeet state boundaries (CC BY 2.5 IN).
Deliverables	Task 1 - data/raw/*.csv json (raw with source columns), data/processed/*.csv (clean), data/data_dictionary.md csv Task 2 - code/06_ml_models.py, model weights outputs/models/*.joblib, metrics outputs/ml/*.csv, ml_summary.json Task 3 - outputs/stats/statistical_summary.md + T1-T10 validation tables, figures F01-F10 Task 4 - blog/blog_post.md html, blog/images, blog/interactive/*.html, blog/engagement (log template, audit report) Task 5 - video/Indian_Festivals_Project_Report.mp4 (1080p, narrated), video/narration_script.md, this report
Approach	1. Scrape the RBI holiday matrix for every office/month/year (2,031 office-date rows), save raw HTML + CSV with source URL per row. 2. Build a 99-festival regex catalog; resolve each RBI label into festivals; split shared-date closures with attribution_weight = 1/k; flag festivals whose reach is not identifiable. 3. Source-ground each festival's tradition from its Wikipedia infobox ('Observed by') or an official Govt. page. 4. Download Census 2011 C-01, map 34 RBI offices to 29 states/UTs, build festival-, observation- and state-level tables. 5. Collect 78 economic/footfall figures; open every page to confirm the figure (snippet-only figures were excluded). 6. EDA & hypothesis tests (chi-square, permutation, Spearman, Kruskal-Wallis, Mann-Whitney, OLS). 7. ML: K-Means/Ward clustering of states, Random Forest vs Logistic tradition classifier, log-linear trade regression with LOO and temporal hold-out. 8. Blog + interactive Plotly charts, engagement audit tooling; 9. narrated video; 10. report.

Code

01_scrape_rbi_holidays.py 02_verify_wikipedia.py 03_economic_footfall_records.py
04_build_clean_dataset.py 05_eda_statistics.py 06_ml_models.py
07_data_dictionary.py 08_interactive_charts.py 09_blog_and_engagement.py
10_make_video.py 11_build_report.py festival_catalog.py viz_style.py

Excerpt - fractional attribution of shared-date holidays (04_build_clean_dataset.py):
obs["n_items_on_date"] = obs.groupby(["date","rbi_regional_office"])["festival"]
.transform("size") + obs["n_civic_items"]
obs["attribution_weight"] = 1 / obs["n_items_on_date"]
obs["attribution"] = np.where(obs["n_items_on_date"] == 1, "unique", "shared_date")

Task 1: Dataset

2,705 festival-holiday observations | 99 festivals | 34 RBI offices | 29 states/UTs | 2024-2026 | 78 verified economic/footfall records | Census 2011 demography. Zero synthetic records.

tradition	festivals	holiday-days/office/yr	% of festival holidays	% moving-date
Hindu	45	7.22	46.5	68.89
Tribal/Indigenous	22	1.13	7.3	63.64
Muslim	7	2.69	17.3	100.0
Christian	7	2.1	13.5	14.29
Buddhist	6	0.63	4.0	83.33
Sikh	5	0.43	2.8	60.0
Cultural (multi-faith)	4	0.51	3.3	25.0
Jain	2	0.37	2.4	100.0
Parsi	1	0.44	2.9	0.0

Limitations: RBI publishes one combined label per date, so co-listed festivals share the day (39 of 99 festivals never appear alone -> reach flagged not identifiable); bank holidays measure official recognition, not participation; trade figures are industry estimates.

Task 3: Statistics

variable	n	mean	median	std	skewness
festival_expected_states_per_year	99.0	4.11	1.27	6.32	2.44
festival_mean_holiday_days (duration)	99.0	1.03	1.0	0.09	4.44
festival_date_drift_days (moving festivals)	64.0	23.19	19.0	43.7	7.65
state_festival_holiday_days_per_year (frequency)	29.0	15.39	15.32	2.49	0.27
state_holiday_diversity_shannon	29.0	1.54	1.52	0.14	0.46
state_distinct_festivals_2024_26	29.0	34.97	35.0	5.68	-0.52
festive_trade_INR_crore (spending)	18.0	118166.67	37500.0	173144.7	1.77

id	test	statistic	p_value	result
H1	Chi-square goodness-of-fit (12 months, weighted)	551.155	< 0.001	Reject H0
H2	Chi-square test of independence (+5,000-permutat	7.719	0.578	Fail to reject H0

	H3-Hindu	Spearman rank correlation (n=29 states/UTs)	0.575	0.001	Reject H0																								
	H3-Muslim	Spearman rank correlation (n=29 states/UTs)	0.422	0.022	Reject H0																								
	H3-Christian	Spearman rank correlation (n=29 states/UTs)	0.532	0.003	Reject H0																								
	H3-Sikh	Spearman rank correlation (n=29 states/UTs)	0.536	0.003	Reject H0																								
	H3-Buddhist	Spearman rank correlation (n=29 states/UTs)	0.709	< 0.001	Reject H0																								
	H3-Jain	Spearman rank correlation (n=29 states/UTs)	0.639	< 0.001	Reject H0																								
	H4	Kruskal-Wallis H (identifiable-reach festivals)	7.251	0.064	Fail to reject H0																								
	H5	Mann-Whitney U	885.500	0.270	Fail to reject H0																								
	H6-hol_days	Kruskal-Wallis H	3.788	0.580	Fail to reject H0																								
	H6-holiday_	Kruskal-Wallis H	4.587	0.468	Fail to reject H0																								
	H7	Spearman rank correlation	0.248	0.194	Fail to reject H0																								
	H8	Mann-Whitney U on log trade (national CAIT-type)	52.000	0.109	Fail to reject H0																								
	H9	OLS log(trade) ~ year + festival fixed effects	12.469	< 0.001	Reject H0																								
	H1: holidays cluster by month (chi2 = 551, p < 0.001; peaks Mar, Oct, Apr). H2: season independent of tradition (perm. p = 0.58). H3: holiday share correlates with population share for all six faiths (rho 0.42-0.71, p < 0.05). H4-H7 not significant. H9: festive trade grows 34.0%/yr (95% CI 27.2 to 41.1).																												
Task 2: Machine Learning	A. Clustering states (K-Means vs Ward) Features: Hindu/Muslim/Christian/Tribal shares of festival holidays (chosen by a documented feature-set experiment). k = 4 (silhouette 0.413); bootstrap ARI 0.61 +/- 0.28; ARI vs region 0.03. C1: above-avg Christian holiday share: Arunachal Pradesh, Goa, Mizoram, Nagaland; C2: above-avg Hindu holiday share: Andhra Pradesh, Assam, Bihar, Gujarat, Himachal Pradesh, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Odisha, Punjab-Haryana-Chandigarh, Rajasthan, Sikkim, Tamil Nadu, Telangana, Uttar Pradesh, Uttarakhand, West Bengal; C3: above-avg Muslim holiday share: Chhattisgarh, Delhi, Jammu & Kashmir, Kerala; C4: above-avg Tribal Indigenous holiday share: Manipur, Meghalaya, Tripura B. Tradition classifier (5-fold x 20 repeated stratified CV) <table> <tr> <th>model</th><th>accuracy_mean</th><th>macro_precision_mean</th><th>macro_recall_mean</th><th>macro_f1_mean</th><th>macro_f1_sd</th></tr> <tr> <td>Majority baseline</td><td>0.455</td><td>0.114</td><td>0.25</td><td>0.156</td><td>0.002</td></tr> <tr> <td>Logistic Regression</td><td>0.434</td><td>0.42</td><td>0.424</td><td>0.4</td><td>0.106</td></tr> <tr> <td>Random Forest</td><td>0.464</td><td>0.419</td><td>0.424</td><td>0.404</td><td>0.105</td></tr> </table> C. Festive-trade regression (log-linear, festival fixed effects) n = 15; in-sample R2 (log) 0.993; LOO RMSE 0.281 log units; LOO MAPE 20.2%; growth 35.5%/yr; hold-					model	accuracy_mean	macro_precision_mean	macro_recall_mean	macro_f1_mean	macro_f1_sd	Majority baseline	0.455	0.114	0.25	0.156	0.002	Logistic Regression	0.434	0.42	0.424	0.4	0.106	Random Forest	0.464	0.419	0.424	0.404	0.105
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	out 2026 MAPE 13.1% (Holi 2.9%, Raksha Bandhan 23.2%).
Task 4: Blog & Engagement	Article 'What India's Bank-Holiday Calendar Reveals About How the Country Celebrates' (~1,700 words, 9 figures, 8 footnoted sources, secular framing) in blog/blog_post.md and a paste-ready blog_post.html; 4 interactive Plotly charts (hover tooltips) in blog/interactive/. Publication status: NOT YET PUBLISHED. External interactions logged: 0 / 40 target; comments answered: 0. The audit (09_blog_and_engagement.py) counts only people outside VIT and requires a screenshot per row. Publishing and community engagement must be done by the author; no engagement is simulated.
Task 5: Video	video/Indian_Festivals_Project_Report.mp4 - 1920x1080, H.264/AAC, 11 narrated slides (~5.4 min): motivation & source integrity, statistics, ML architecture & performance, economics, blog & reception, takeaways. The reception slide reads engagement_metrics.json; re-run 10_make_video.py after publishing to produce the final cut.
Output Visuals	Source: Reserve Bank of India holiday matrix 2024-26 (rbi.org.in). Shared-date holidays split equally among co-listed items. Fig 1. Festival bank-holiday days by month and tradition (RBI 2024-26). State-wise festival heatmap: when each state's holidays fall Fig 2. State x month heatmap. Source: Reserve Bank of India holiday matrix 2024-26 (rbi.org.in)

Festival bank holidays per year, by state

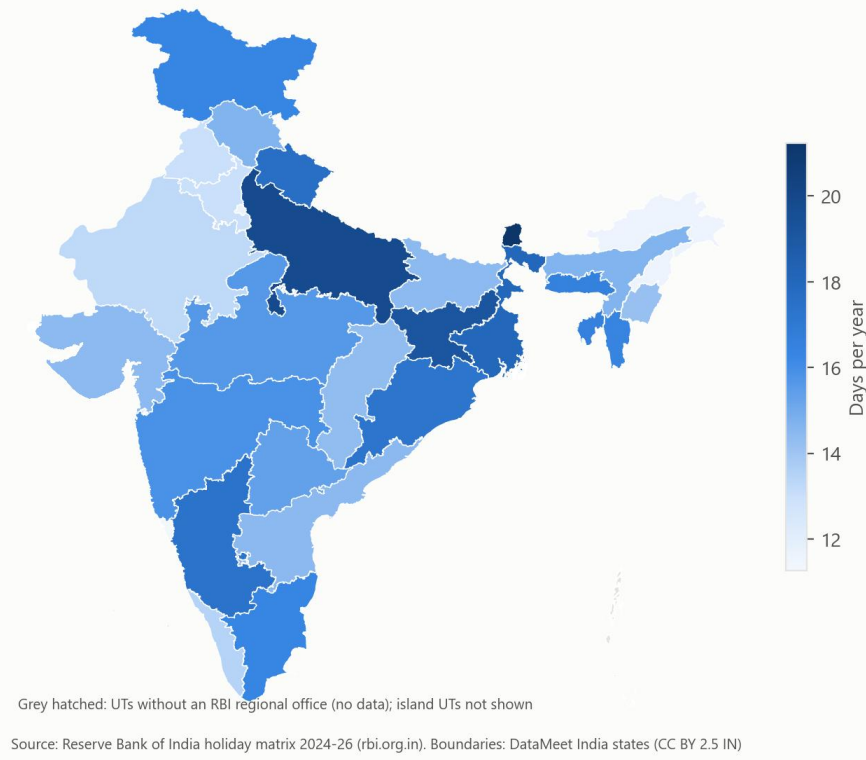


Fig 3. Festival holidays per year by state.

Do holiday calendars mirror demography? (each dot = one state; dashed = trend)

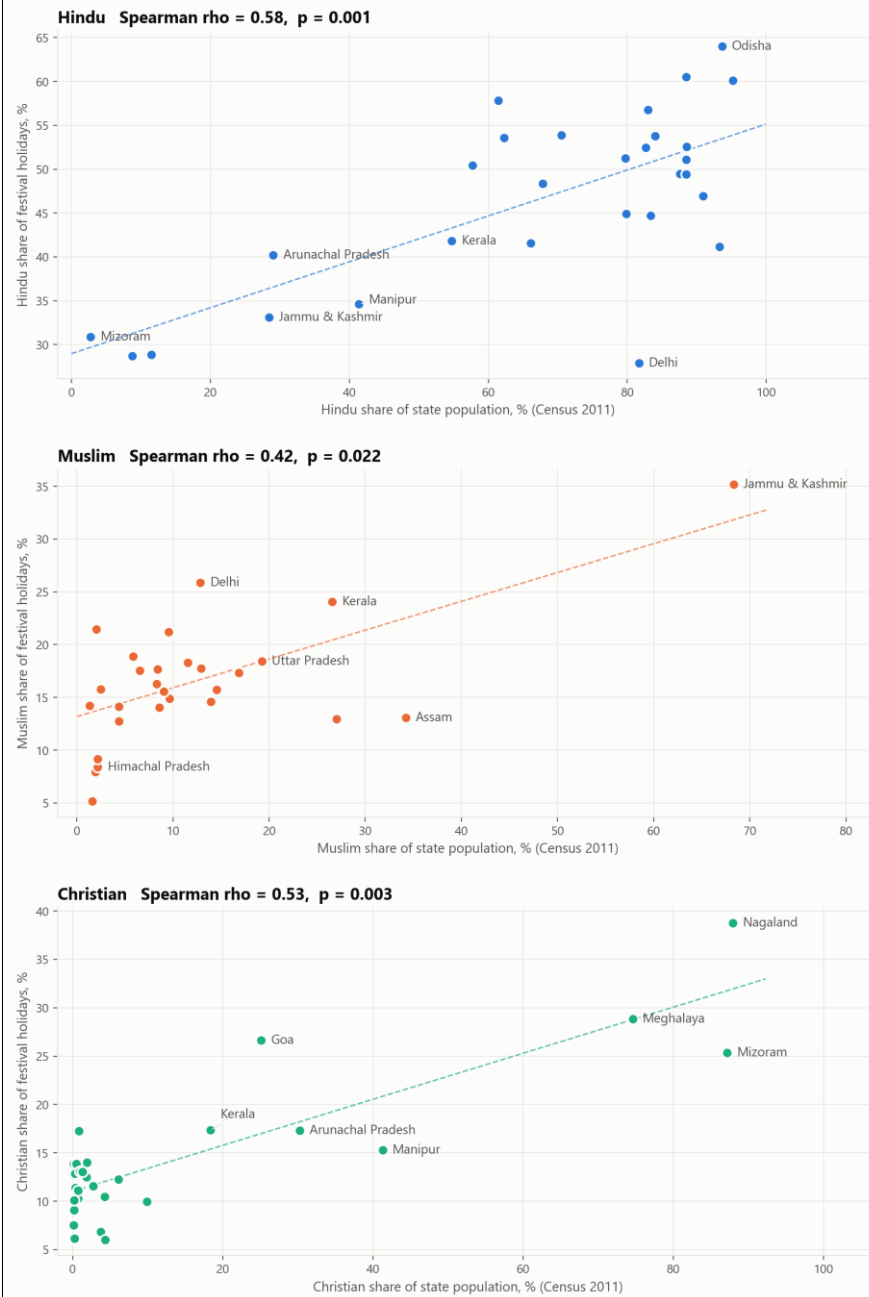


Fig 4. Holiday share vs Census population share.

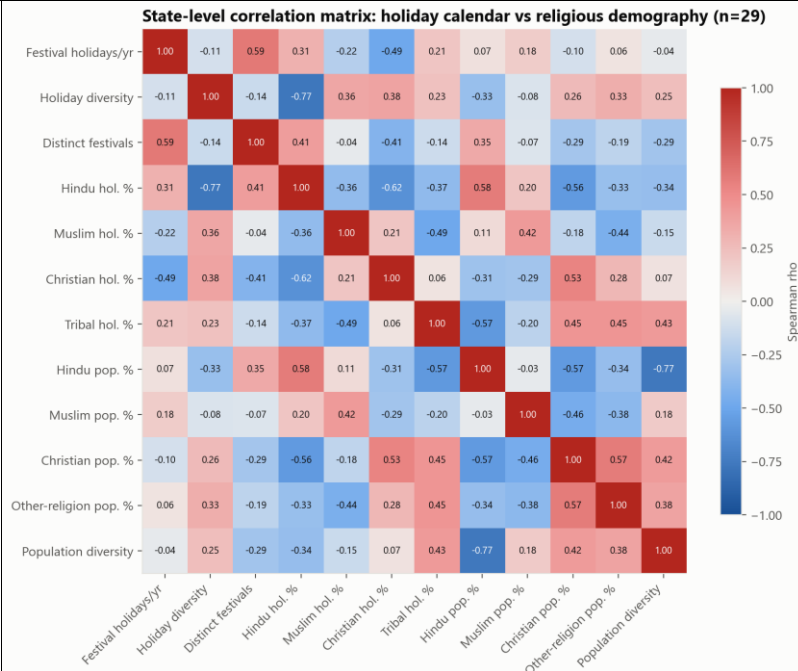


Fig 5. State-level Spearman correlation matrix.

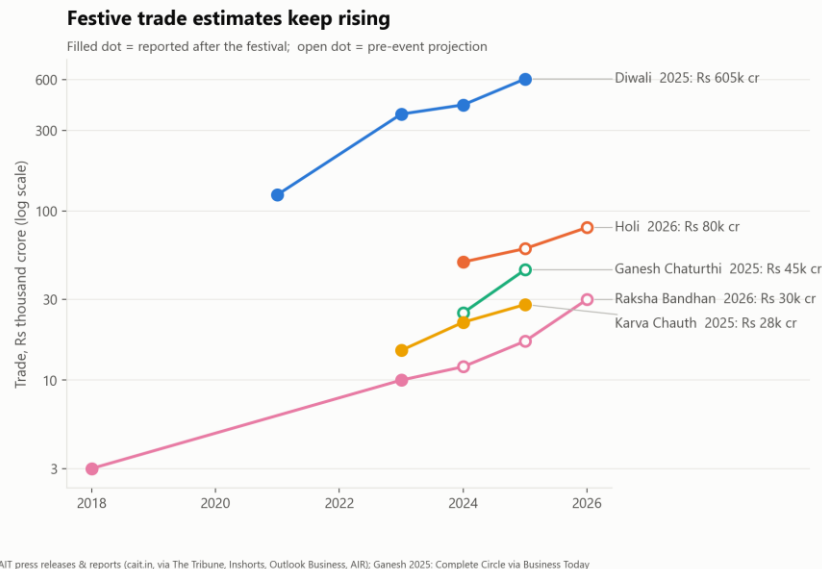


Fig 6. Festive trade estimates (CAIT).

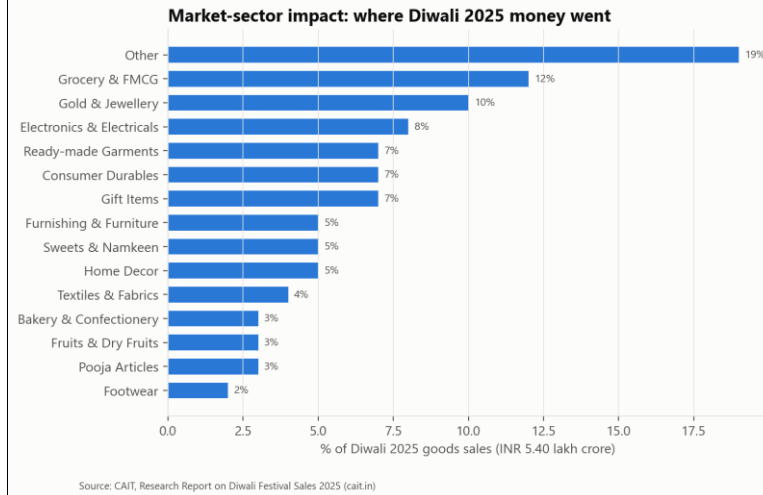


Fig 7. Diwali 2025 sector split.

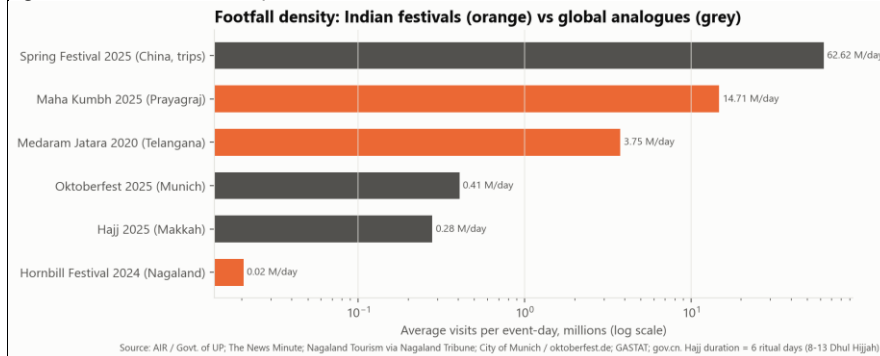


Fig 8. Footfall density vs global analogues.

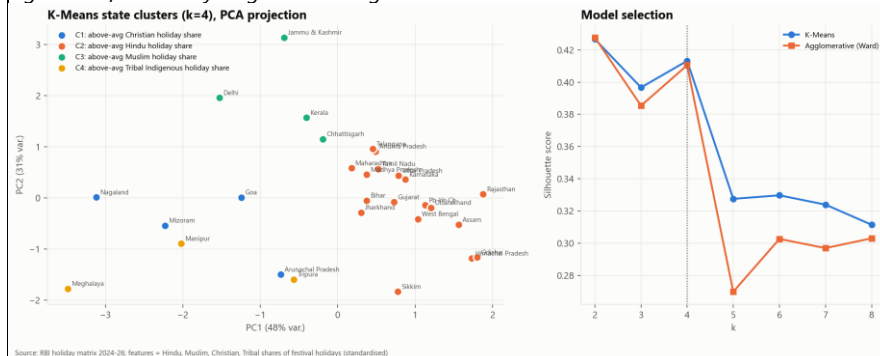
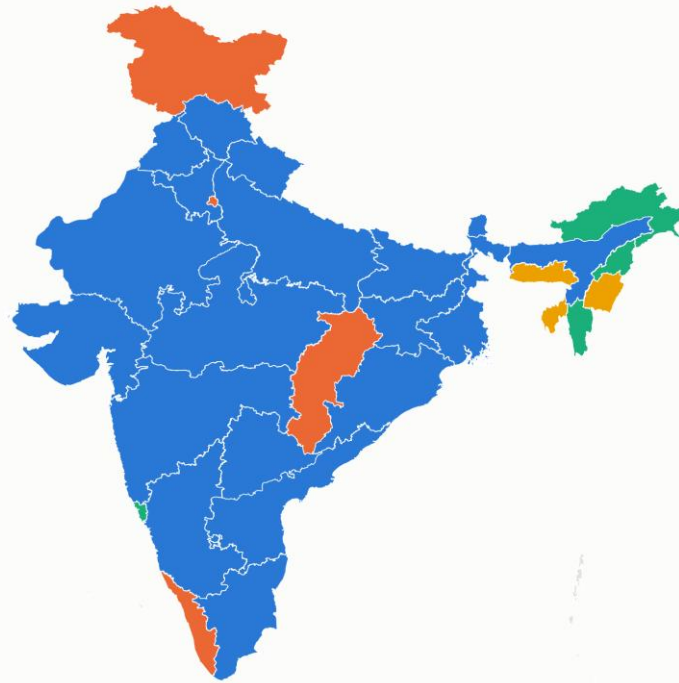


Fig 9. K-Means state clusters (PCA) and model selection.

Four kinds of festival calendar in India

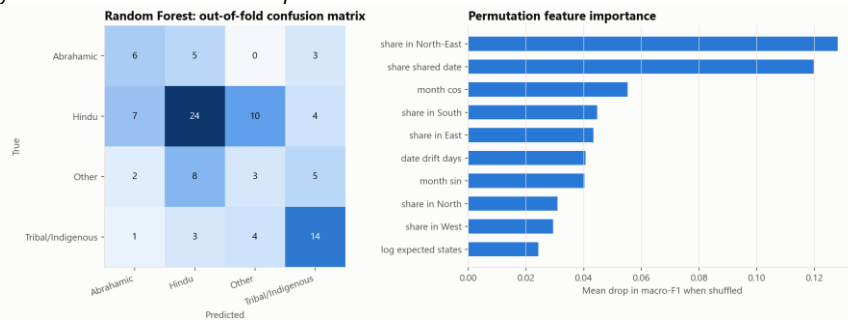
States grouped by the share of Hindu, Muslim, Christian and tribal festivals in their bank holidays (K-Means clustering)



- More Hindu holidays than average (18 states)
- More Christian holidays than average (4 states)
- More Muslim holidays than average (4 states)
- More Tribal/indigenous holidays than average (3 states)
- No RBI office (no data)

Source: RBI holiday matrix 2024-26; boundaries DataMeet (CC BY 2.5 IN); island UTs not shown

Fig 9b. State clusters on the map.



Source: Festival features derived from RBI holiday matrix 2024-26; labels from Wikipedia / Govt. attribution (festival_mastereco)

Fig 10. Classifier confusion matrix and permutation importance.

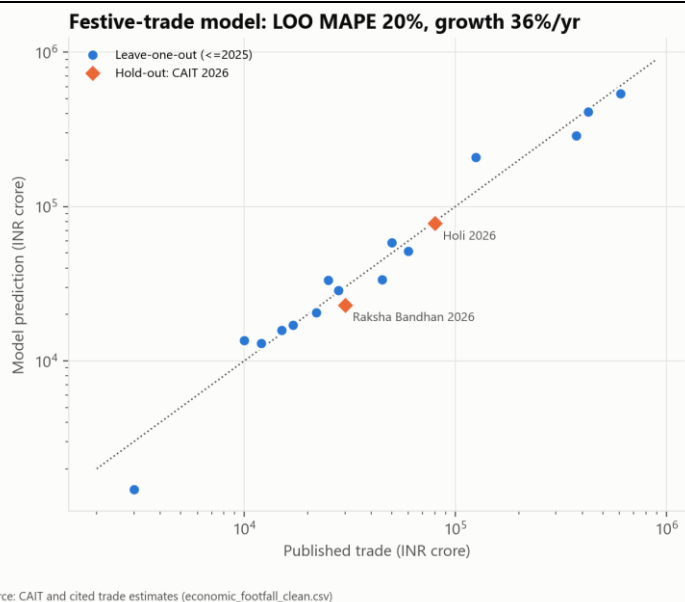


Fig 11. Trade model: LOO and 2026 hold-out predictions.

Insights	1. Festival holidays peak in March (2.7 days/office), October (2.5) and April (2.1); February and July are quietest. 2. Every faith's holiday share rises with its population share (Spearman 0.42-0.71), yet the calendar is more plural than the population: Hindu festivals are ~46% of festival holidays vs 79.8% of population, because Christmas, Good Friday, both Eids, Buddha Purnima, Mahavir Jayanti and Guru Nanak Jayanti are gazetted almost nationwide. 3. Sikkim (21.2), Uttar Pradesh (19.8) and Jharkhand (19.1) have the most festival holidays; Goa (11.3) the fewest. Regions do not differ significantly. 4. States cluster by calendar culture, not geography (ARI vs region 0.03). 5. A festival's date and footprint only weakly reveal its faith (macro-F1 0.40): nationwide festivals of all faiths look alike. 6. Reported festive trade grows ~34%/yr (Diwali: Rs 1.25 lakh cr in 2021 -> Rs 6.05 lakh cr in 2025), but national spending estimates exist mainly for Hindu festivals - a measurement blind spot.
Conclusion	A fully source-grounded, multi-faith festival dataset was built from official RBI and Census data plus 78 verified economic figures, analysed with hypothesis tests and three ML models, and communicated through a footnoted blog, interactive charts and a narrated video. Remaining author actions: publish the article, collect 40+ genuine external interactions, fill the engagement log, and re-run the audit and video scripts.
References	[1] RBI, Holidays under NI Act - https://www.rbi.org.in/Scripts/HolidayMatrixDisplay.aspx [2] Census of India 2011, C-01 - https://censusindia.gov.in/nada/index.php/catalog/11361 [3] CAIT Diwali 2025 report - https://cait.in (10 Nov 2025); other economic sources listed per record in data/raw/economic_footfall_raw.csv [4] All India Radio, Maha Kumbh concludes - https://www.newsonair.gov.in/maha-kumbh-mela-2025-concludes-in-prayagraj-on-maha-shivratri [5] NRF (16 Jan 2025); gov.cn (5 Feb 2025); GASTAT (5 Jun 2025); muenchen.de - global analogues [6] DataMeet India state boundaries (CC BY 2.5 IN) - github.com/datameet/maps